

FAILSAFE

AI-Powered Early Student Risk Detection and Intervention System

Technical Report — Machine Learning, Explainability & LLM Integration

Dataset: UCI Student Performance Dataset — Model: XGBoost + SHAP

1. Problem Statement and Motivation

In many educational institutions, academic failure is identified only after end-semester evaluations — far too late for any meaningful intervention. Faculty members lack intelligent, data-driven tools to detect struggling students early, understand the root causes of their difficulties, or take proactive corrective action.

FAILSAFE addresses this gap by combining a trained machine learning model, SHAP-based explainability, and LLM-generated intervention plans into a deployable web system. The key design goals are:

- Predict student failure risk *before* final grades, using attendance, grades, behavioral patterns, and support indicators.
- Explain each prediction transparently so non-technical faculty can act on the results.
- Auto-generate personalized, actionable intervention plans per student.
- Provide a faculty-facing dashboard for monitoring and tracking at-risk cohorts.

2. Dataset and Feature Engineering

2.1. Dataset

The system is trained on the **UCI Student Performance Dataset** (Portuguese secondary school students), using the Portuguese language course file (`student-por.csv`, $n = 649$). The dataset includes academic records, attendance, family background, support indicators, and demographic features.

The binary target variable **risk** is defined as:

$$\text{risk} = \begin{cases} 1 & \text{if } G3 < 10 \\ 0 & \text{otherwise} \end{cases}$$

Class imbalance (approximately 85% low-risk, 15% high-risk) is addressed using **SMOTE** during training.

2.2. Features Used

After dropping low-signal columns (school, reason, guardian, travel time, social habits), the following features are retained:

Table 1: Input Features

Feature	Description
G1, G2	Period 1 and Period 2 grades (0–20)
absences	Total recorded absences
failures	Number of past class failures
studytime	Weekly study time (1=<2h to 4=>10h)
Medu, Fedu	Mother’s and father’s education level (0–4)
age	Student age
sex, address	Demographic indicators
schoolsup, famsup, paid	Support availability flags
activities, nursery, higher	Extracurricular and aspiration indicators

2.3. Engineered Features

Five custom features are constructed to capture hidden academic risk patterns:

Table 2: Engineered Features

Feature	Formula / Rationale
grade_drop	$G2 - G1$: detects downward academic trajectory
failure_x_nosupport	$\text{failures} \times (\text{no school support} + \text{no family support})$: compound risk
G1_x_G2	interaction term capturing joint grade performance
support_deficit	count of missing support sources (school, family, paid classes)
parent_edu_index	$\text{Medu} + \text{Fedu}$: parental education combined score
disadvantage_score	$\text{low parent education} + \text{support deficit}$: socio-academic vulnerability

3. Machine Learning Pipeline

3.1. Preprocessing

A `ColumnTransformer` applies three transformers in parallel:

- **Numeric columns:** `StandardScaler`
- **Absences:** Log transform (`np.log1p`) followed by `StandardScaler` (right-skewed distribution)
- **Binary categorical columns:** `OneHotEncoder` with `drop='first'` (sex, address, support flags)

3.2. Models Evaluated

Six classifiers were benchmarked with the same preprocessing pipeline and SMOTE:

Table 3: Model Comparison

Model	Accuracy	Precision	Recall	F1 (Risk Class)
Logistic Regression	0.88	0.71	0.65	0.68
K-Nearest Neighbours	0.85	0.62	0.60	0.61
Support Vector Machine	0.89	0.73	0.70	0.71
Decision Tree	0.86	0.65	0.70	0.67
Random Forest	0.92	0.80	0.75	0.77
XGBoost (final)	0.94	0.77	0.85	0.81

XGBoost was selected as the final model due to its superior recall on the minority class (at-risk students), which is the operationally critical metric — missing a high-risk student is more costly than a false alarm.

3.3. Hyperparameter Tuning

RandomizedSearchCV with **StratifiedKFold** (5 folds) was applied over the following search space:

```
n_estimators: [100, 300, 500]  max_depth: [3, 4, 5]  learning_rate: [0.001, 0.01, 0.05, 0.1]
subsample: [0.7, 0.8, 1.0]  colsample_bytree: [0.7, 0.8, 1.0]  gamma: [0, 0.1, 0.5]
```

Best parameters found: `n_estimators=500`, `max_depth=3`, `learning_rate=0.001`, `gamma=0.5`, `subsample=1.0`, `colsample_bytree=0.7`

Best CV Score: 0.825 (stratified F1)

3.4. Final Model Performance

Table 4: Classification Report — XGBoost (Test Set, $n = 130$)

Class	Precision	Recall	F1-Score	Support
Low Risk (0)	0.97	0.95	0.96	110
High Risk (1)	0.77	0.85	0.81	20
Weighted Avg	0.94	0.94	0.94	130

The model achieves **85% recall** on at-risk students, correctly flagging 17 out of 20 true failure cases in the test set — the primary success criterion for an early-warning system.

4. Explainability with SHAP

4.1. Approach

SHAP (*SHapley Additive exPlanations*) **TreeExplainer** is applied to the XGBoost component of the pipeline after preprocessing. For each student prediction, the top-5 features by SHAP magnitude are extracted and presented as human-readable risk factors.

4.2. Key SHAP Insights

Top risk-increasing features (positive SHAP):

- **Low G2** — the strongest single predictor; a declining Period 2 grade is the clearest early warning signal.
- **Low G1** — consistent with G2, base academic performance strongly drives risk.
- **High failure_x_nosupport** — students with past failures *and* no institutional or family support are disproportionately at risk.
- **High grade_drop** — a significant G1→G2 decline indicates an emerging downward trajectory even when absolute grades are not yet critical.
- **High absences** — strong secondary predictor; chronic absenteeism correlates with disengagement.

Top risk-reducing features (negative SHAP):

- Higher Edu aspiration (**higher=yes**) — students aiming for university education show significantly lower failure probability.
- High G1 × G2 interaction — strong and consistent grades across both periods act as protective factors.
- Low support_deficit — access to paid classes, school support, and family support collectively reduces risk.

4.3. Explainability in Practice

Each prediction served through the API includes a structured explanation:

Listing 1: Sample SHAP output for a high-risk student

```
{
  "risk_tier": "HIGH",
  "risk_prob": 82.4,
  "shap_reasons": [
    {"label": "Period 2 Grade",      "direction": "increases risk", "shap": 0.612},
    {"label": "Grade Drop (G2-G1)",  "direction": "increases risk", "shap": 0.481},
    {"label": "Failures x No Support", "direction": "increases risk", "shap": 0.394},
    {"label": "Absences",             "direction": "increases risk", "shap": 0.287},
    {"label": "Higher Edu Goal",      "direction": "reduces risk",   "shap": -0.210}
  ]
}
```

This output is rendered as animated horizontal bar charts in the frontend, color-coded red (risk-increasing) and green (risk-reducing) for faculty readability.

5. System Architecture and Key Features

5.1. Architecture Overview

Table 5: Technology Stack

Layer	Technology	Role
ML	XGBoost, Scikit-learn, SHAP	Model training, preprocessing pipeline, explainability
Imbalance	Imbalanced-learn (SMOTE)	Synthetic oversampling of minority class
Backend	FastAPI + Uvicorn	REST API, prediction serving, LLM orchestration
LLM	Groq API (Llama 3.3 70B)	Generates personalized intervention recommendations
Frontend	HTML5, CSS3, Vanilla JS	Faculty dashboard, form input, results visualization
Storage	Joblib (.pkl)	Serialized model pipeline

5.2. API Endpoints

POST /predict — single student prediction. Accepts 16 student attributes, returns:

- **risk_tier** (HIGH / MEDIUM / LOW) with threshold: ≥ 0.65 HIGH, ≥ 0.35 MEDIUM
- **risk_prob** as a percentage
- Top-5 SHAP explanations with direction labels
- LLM-generated intervention plan (if API key is configured)

POST /predict/bulk — CSV upload for class-level analysis. Returns per-student risk tiers and a cohort summary (total / high / medium / low counts).

5.3. Frontend Features

The faculty-facing dashboard (pure HTML/CSS/JS, no framework dependency) includes:

- **Single Student View:** 16-field input form with dropdowns and number inputs, risk badge with animated probability bar, SHAP waterfall-style bar chart, and LLM intervention card.
- **Bulk Upload View:** CSV drag-and-drop, cohort summary statistics panel (4 stat boxes), filterable student results table sortable by risk tier.

- Dark industrial aesthetic with monospaced typography and color-coded risk indicators (red / amber / green).

5.4. LLM Intervention Generation

When an Anthropic/Groq API key is configured, the system sends a structured prompt to an LLM containing the student’s risk tier, probability, top SHAP factors, and key metrics (G1, G2, absences, failures, study time). The LLM returns 3 specific, actionable intervention recommendations such as:

- Weekly one-on-one mentoring sessions with subject faculty
- Referral to counselling services for attendance and motivation issues
- Enrolment in structured remedial study groups
- Parental communication regarding support deficit

6. Risk Stratification

Student predictions are mapped to three actionable tiers:

Table 6: Risk Tier Definitions

Tier	Probability	Recommended Action
HIGH	$\geq 65\%$	Immediate faculty intervention, counselling referral, attendance monitoring
MEDIUM	35%–64%	Scheduled check-in, study plan review, support resource signposting
LOW	$< 35\%$	Routine monitoring, no immediate action required

7. Conclusions and Future Work

FAILSAFE demonstrates that combining classical ML (XGBoost + SMOTE), explainability (SHAP), and generative AI (LLM intervention plans) into a unified, deployable system is both technically feasible and practically valuable for educational institutions.

Key outcomes:

- 94% overall accuracy and 85% recall on the at-risk minority class
- Transparent, per-student SHAP explanations accessible to non-technical faculty
- End-to-end deployable system (FastAPI backend + vanilla JS frontend) with no cloud dependency beyond optional LLM

Planned extensions:

- Faculty authentication and role-based access control
- Longitudinal student tracking and semester-over-semester risk trend visualisation
- LMS integration (Moodle, Canvas) for automated data ingestion
- Real-time alert system with email/SMS notifications for high-risk flags
- Multi-institutional deployment with institution-specific model fine-tuning